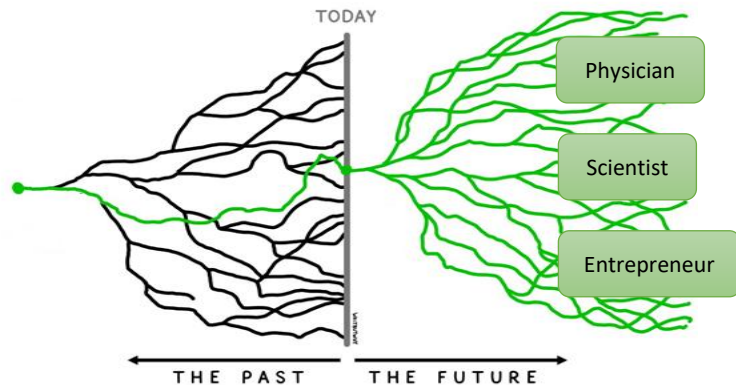


LEADERSHIP CONTEXT STATEMENT

Overview: Path to Today

In medical school, I approached each rotation with an open mind, envisioning myself in every specialty. Initially I was drawn to surgery which led me to travel to the United States for my elective rotations. In addition to the 4-week rotations I did at UC and UIC, I did an 8-week rotation at WashU with Dr. William Hawkins, chief of the surgery department. Dr. Hawkins recognized my strong work ethic, determination, and thirst for knowledge, so he offered me a position as a postdoctoral research fellow in his lab, which focused on studying new cancer therapies. Despite my limited research experience at the time, he took a chance on me, and I quickly proved myself as a successful researcher, eventually leading a team of scientists on my own project. When the project extended beyond the initial 2-year plan, I chose to extend my time in the lab committed to seeing it through to completion. After 6 years of dedicated work, we uncovered the mechanism of action of a novel small molecule used in targeted therapies for various cancers, including synovial, ovarian and pancreatic cancers (see [Appendix 1](#) reflection about my research experience).



Current role: Benefits & Challenges

Throughout this journey, I developed the skills necessary to lead a research project from its inception through literature review, hypothesis generation, grant funding, experimentation, and publication to peer-reviewed journals. Consequently, I was approached by a startup, QuantumNextDX, to join their leadership as a consultant and serve as their subject matter expert to identify new opportunities and address challenges in pancreatic cancer treatment with the power of quantum computing.

The team is small and very collaborative. It consists of (1) President, CEO, & Co-founder [my mentor], (2) VP, CSO¹, & Co-founder, (3) VP, Director of Purpose & Impact, (4) Consultant [me], and (5) a research assistant. Our collaboration has been incredibly exciting. I get the utmost level of autonomy where I get to design and pursue entire projects from the bottom up. I am fortunate to have their full support in pausing my responsibilities while I complete my clinical training. However, working for my mentor blurs some lines. I am extremely appreciative of the constant help and support I am receiving on all levels, yet we have not yet signed any contracts, nor am I being compensated for my work. So, for the time being, I intend to keep to myself whatever research ideas or hypotheses I am working on and plan on sharing them and building on them only after those lines have been clarified.

¹ Chief Scientific Officer

Professional Goals in the next 5 years

Being at a crossroad prompted a period of self-reflection, allowing me to revisit what brings me the greatest fulfillment.

I concluded that my *raison d'être* is to provide for my family,
by doing what I love, while making a positive impact on the world.

I aim to achieve this by pursuing a career devoted to alleviating the suffering. There is not one but three professional goals, which I will be happy doing one or the other or some mix-match of two at the same time.

Physician. I aspire to become a board-certified physician. This has been my journey for the last thirteen years and I would like to get to the finish line. I can alleviate suffering by delivering compassionate care to patients, particularly during their most vulnerable moments. I am deeply committed to guiding them through illness, forming meaningful connections, and advocating for their well-being. Despite my initial interest, I realized that surgery was not the right path for me. Ideally, I would like to pursue Internal Medicine residency followed by Hematology & Oncology fellowship to practice as an oncologist. However, I also see myself as a primary care physician. I value taking the time to engage with patients on a personal level, prioritize preventive care, and manage chronic conditions. Additionally, I believe in educating patients about their diseases because they, in turn, share this knowledge with their family and friends, becoming ambassadors of their condition.

Scientist. Regardless of ending up as an oncologist, PCP, or neither, I see myself continuing my research endeavors in some shape or form. My postdoctoral experience made me realize the immense potential to bridge scientific discoveries with clinical advancements and its effects on improving patient outcomes. I aspire to lead efforts in developing innovative treatment strategies and offer renewed hope to those affected by diseases refractory to conventional therapies, like pancreatic cancer. Be it with QNDX or in a more established drug development company.

Entrepreneur. I witnessed the complexities of the healthcare industry, and I recognized the importance of complementing my medical expertise with a solid understanding of business principles. I am pursuing this executive MBA to learn the necessary skills to navigate the intricacies of starting a drug development company or the regulations of getting a drug from preclinical trials in the lab to clinical trials then market. Depending on where my research takes me, I can either start the drug company, or start a contract research organization. Much is unknown on how I will utilize my EMBA, but I am very excited about all those possibilities.

Essential Strengths & Hindering Weaknesses for Reaching Goals

I love working with smart people on solving difficult problems. I pride myself on having great leadership skills, communication & listening skills, viewing things from their perspective, and being a lifelong learner that's always looking for the new intellectual challenge. All of these enable me to effectively collaborate with/lead interdisciplinary teams. Additionally, my ability to adapt and innovate in the face of setbacks has been crucial in my research journey.

My unconventional path can discourage some residency programs to accept me into their training program. I have been out of the hospital for more than 5 years now. I am hoping to find the ones that look at me holistically and appreciate my accomplishments.

I like to take informed well-contemplated decisions. While this has its huge upsides, it can sometimes lead to hesitancy, slow down progress, or delay decision-making. This will be challenging when having to take decisions in a fast-paced environment with incomplete variables and high levels of uncertainty.

Moreover, I will require the highest levels of discipline and effective time management which can allow me to work on multiple projects and be there for my family and friends. Learning to be more comfortable in delegating tasks will be a crucial skill to avoid burnout.

Core Values I Strive to Embody as a Leader/Manager



Integrity & Honesty are the foundation of trust and accountability, both in leadership and teamwork. This involves transparency in decision making process and consistently doing the right thing, even when it's difficult or when no one is watching.



Selfless service to others as a doctor by providing compassionate, respectful, & empathetic care, and as a leader, by prioritizing the growth and development of my team and empowering them to achieve their goals.



Seeking excellence, which is not an end goal, but an ongoing process set by being precise, detail-oriented & striving for continuous improvement. Outstanding work or results is inevitable.



Balancing work and life responsibilities by offering flexible work arrangements, fair workload, and respecting time off and encouraging them to recharge with family and friends. This ensures they return to work more focused, motivated, and capable of giving their best.



Writing narrative-driven documents rather than slideshows ensures ideas are carefully structured, well-developed, and deeply contemplated. This elevates discussions and facilitates better decision-making by focusing on truth-seeking & identifying weaknesses. It also prevents masking of incomplete or unclear thinking behind bullet points, unlike slideshows, which are sales tools designed to persuade rather than encourage in-depth analysis

Appendix 1: Reflection on my research experience

Science is frustrating

08/2018, I began my research journey investigating Sigma-2-ligands (S2L), small molecules that bind to Sigma-2 receptors, which are overexpressed on rapidly dividing tumor cells. While S2L was known to selectively kill cancer cells, the mechanism remained elusive. My plan was to study this for 2 years, but science had other plans.

One year in, frustration hit hard. I wrote an email to my PI. Nothing seemed to be working. “I hate this project. I’m chasing after a receptor that’s been studied for 40 years. Why is it overexpressed? What does it do? I don’t think I’m smarter than those who’ve studied this for decades. I’m definitely not smart enough to figure this mystery out in one year. I want to see if I can work on something else.” I never sent that email. It’s still saved in my drafts—a moment of doubt. I decided to keep at it.

1st extension. Year 3: COVID hit, halting almost an entire year of lab work. To continue social distancing, I started working nights in an empty lab. I adapted. My PI called me into his office to teach me an important science lesson: knowing when to quit. His previous postdoc had spent 3 years on this & after picking up where she left off, I was approaching the 3-year mark with no end in sight. However, one of our collaborators saw potential in the model I had developed and convinced my PI to let me continue.

2nd extension. Year 4: I had a decent story, but much remained unexplained. Time was against me & the manuscript was already on Version 8_FinalFinalFinalFINAL. 04/2023, I submitted my manuscript to Molecular Cancer. It was rejected at the editorial level. I tried again 2 weeks later to another journal. Another rejection. 08/2023, we submitted the manuscript to (Nature Cell Death & Disease). They gave us some revisions. To address those...

I extended again. Year 5: In a completely unexpected twist, one of the experiments suggested by the reviewers yielded surprising results. I spent the next 8 months rethinking our hypotheses, testing, adjusting, & ultimately developing a new model to explain these findings. It led to a breakthrough—S2L was increasing LDL uptake in cancer cells, causing cholesterol accumulation and inducing ER stress, which ultimately resulted in cell death. These findings opened new doors for exploring S2L as a therapeutic strategy for various cancers.

Without the insights & support from my PI, collaborators, & even the reviewers, this breakthrough would never have been possible. That's why science thrives on teamwork - our collective expertise is what ultimately drives progress. We submitted version 12 almost a completely new story. The reviewer's comment was brief but powerful: “Very good!” Finally, 04/2024, the manuscript was accepted (<https://www.nature.com/articles/s41419-024-06693-8>)

Science is slow. Science is collaborative. Science is frustrating...yet deeply rewarding.

